

Mruthulan Senthil Nathan

SOFTWARE ENGINEERING | FULL-STACK | AI ENGINEERING INTERN

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Available for a year-long internship in 2027/2028

EDUCATION

Diploma in Information Technology, Singapore Polytechnic | Apr 2025 - May 2028 (Expected)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js / Express, FastAPI

Data and Engineering: PostgreSQL, MySQL, SQLite, REST APIs, JWT and bcrypt authentication, Git/GitHub, automated testing, Render

AI Products: interfaces and consent flows for AI-assisted products; a Singaporean text-to-speech voice (with a teammate)

SELECTED PROJECTS

SignalBridge

Youth support platform with consent and human handoff

Champion, Dell InnovateDash 2026 | Featured by Singapore Polytechnic

- Built the youth-facing chat, sign-in and dashboard, and the consent step where the young person previews the note before it is shared.
- Wrote the FastAPI routes that connect the chat to the backend, and the Discord integration with tests for its intake (Next.js, FastAPI, PostgreSQL).
- Team of four Year 2 IT students, for a brief from Singapore Children's Society.

Boss Breaker

Full-stack wellness game, solo build | github.com/mru34/bedca2

Back-end development module coursework

- Built it alone: a nine-page frontend (dashboard, challenges, boss raid, shop, inventory), an Express API and a nine-table MySQL schema created at start-up.
- Points and boss damage are worked out on the server so the browser cannot cheat; bcrypt-hashed passwords and JWT-protected routes.

MEANT

On-device AAC assistant that suggests replies; the user picks every word

2nd Runner-Up, polytechnic category (S\$3,000), Dell InnovateFest 2026 | Featured in Tamil Murasu

- Designed the UI and interaction flows: boards readable at a glance (a colour and an openly licensed symbol per kind of word), search across the whole vocabulary when a prediction is wrong, and setup that stays out of the conversation.
- Worked on the Singaporean text-to-speech voice with a teammate, and presented the build.

Better Call Bhai

Booking site for a Singapore barbershop

Deployed pilot with the shop

- Built the booking flow: customers pick a service, a day and an open slot, then confirm on WhatsApp.
- Node/Express API on SQLite with server-side rules (no past dates; each slot booked once) and an admin page for the shop; deployed on Render.

KnowCad

Retrieval-based AI assistant for customer service

Champion, Autodesk Singapore Hackathon 2026

- Built and delivered the winning presentation for a mixed team; Autodesk engineers led the code.

EXPERIENCE AND LEADERSHIP

- **Retail Associate, Watsons Singapore** | Mar - Apr 2025 - Served customers, processed payments, supported self-checkout and replenished stock in a fast-paced retail environment.
- **Secretary, Subcommittee, Youth Harmony Chapter** - Coordinated communication and follow-through for student activities and community engagement.
- **Class Chairman and SP ACER, Singapore Polytechnic** - Led class communication and supported outreach, including Open House 2026 and First Steps with SP.